NextAIRE

NextAIRE

Next Generation AI Researchers for Air Quality Excellence

Funded by
the European Union

NextAIRE

NextAIRE

Next Generation AI Researchers for
Air Quality Excellence

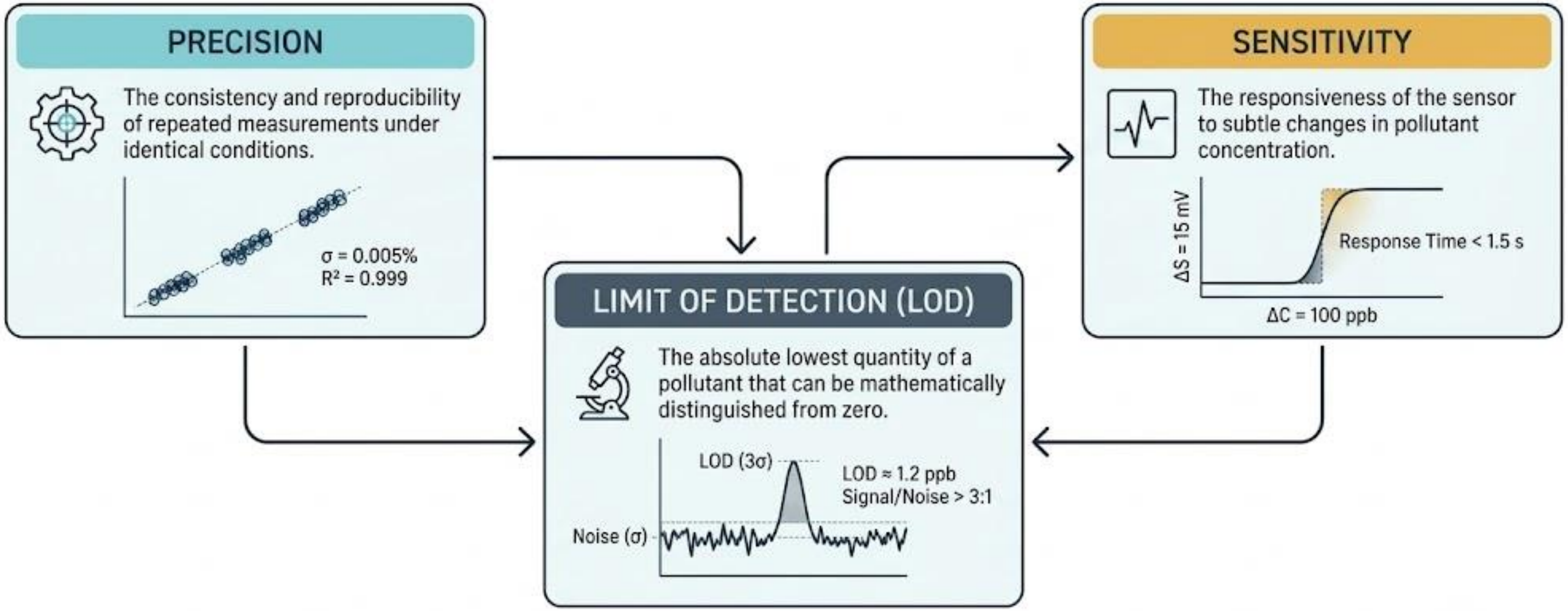
Funded by
the European Union

NextAIRE

Air quality sensors

Funded by
the European Union

Three core metrics separate actionable dana from environmental noise

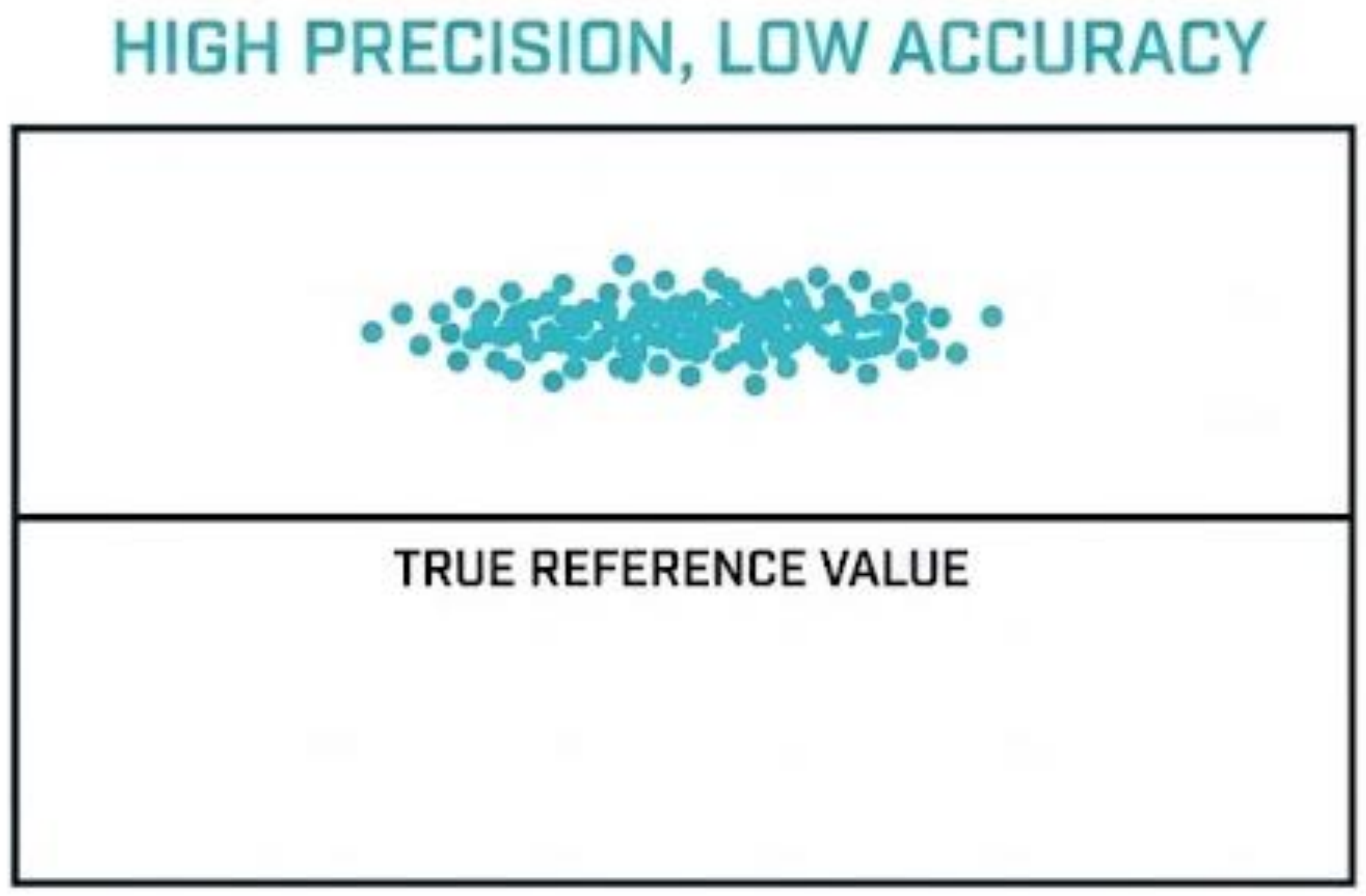


Overall device performance relies on the synthesis of these three metrics alongside calibration stability and long-term reliability.

CALIBRATION STABILITY	$\Delta t < 0.1\%/yr$
LONG-TERM RELIABILITY	MTBF > 50,000 hours

Funded by the European Union

Precision measures consistency independently of true accuracy

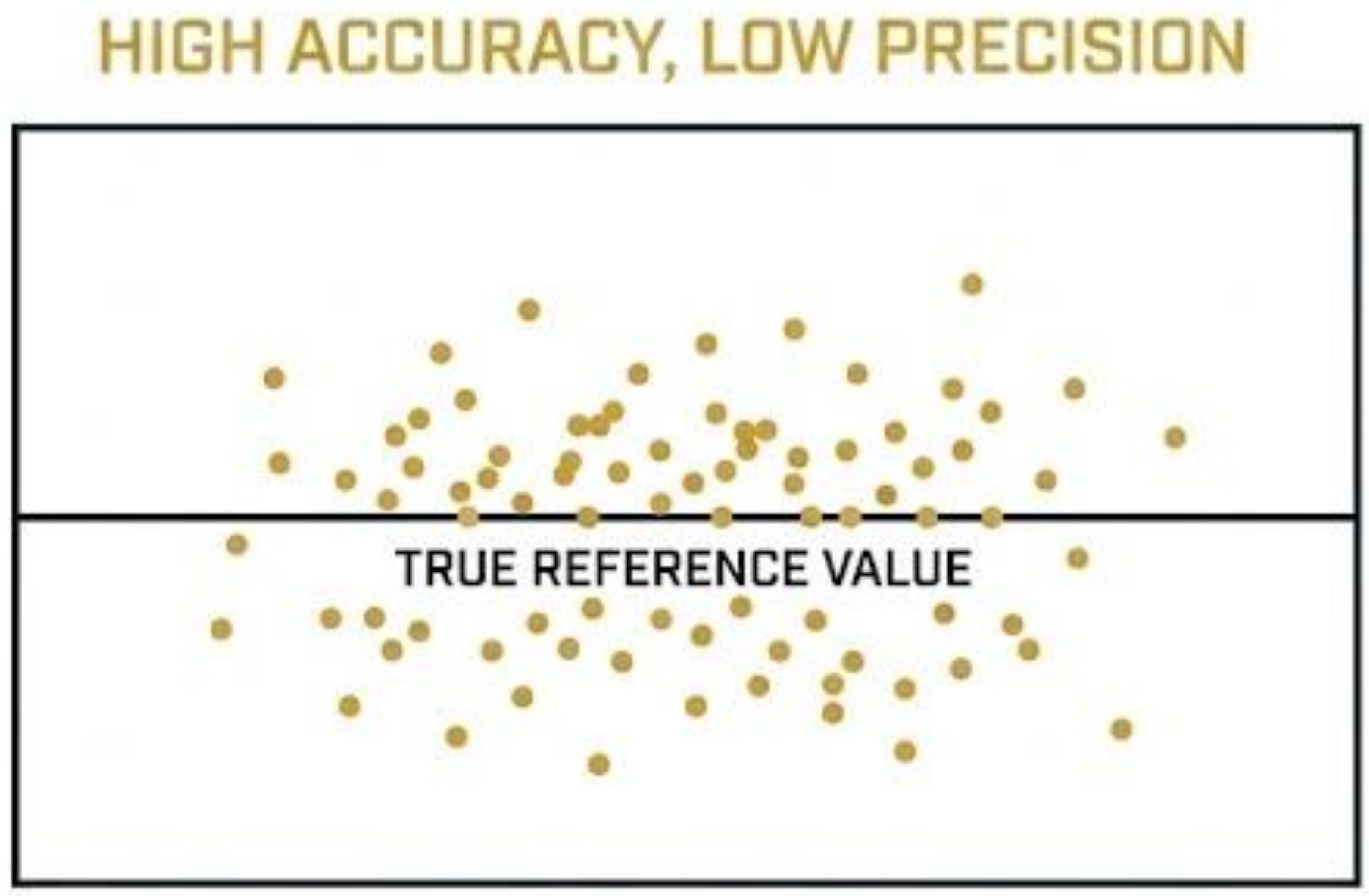


Precision

- Examines the drift between subsequent readings
- The closeness of readings under identical conditions determines measurement precision
- Crucially, precision tests do not take the true refernce value into account

Funded by
the European Union

Precision measures consistency independently of true accuracy



- **Accuracy**
  - The closeness of the measured value to the actual value
  - Exposing a sensor to 4000 ppb SO_2 gas – an output of 4002 ppb is highly accurate, 4300 ppb is not

Funded by
the European Union

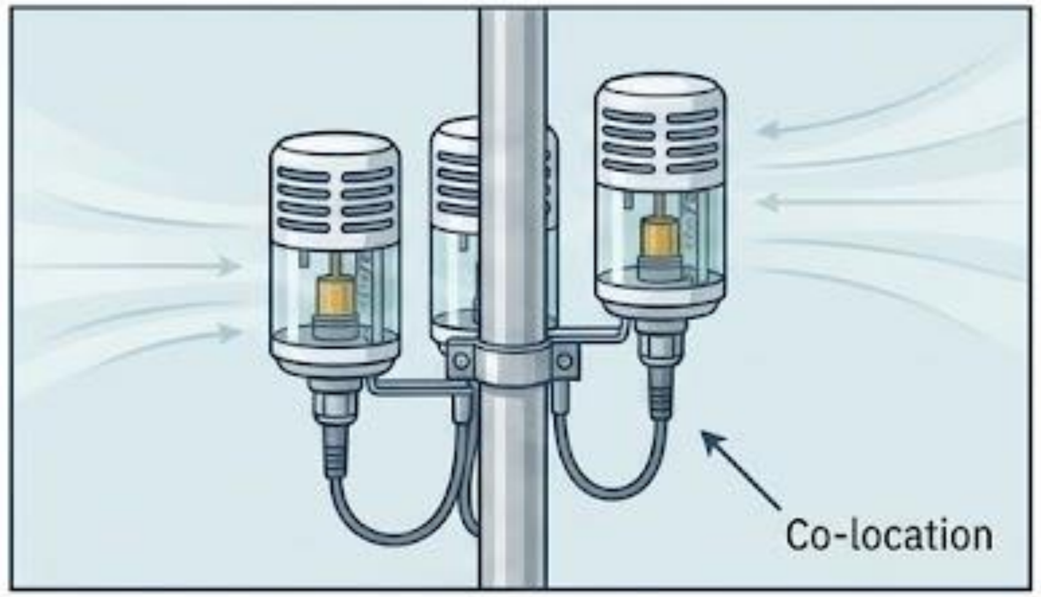
Quantifying precision requires reproducibility analysis and the coefficient of variation

CV: Coefficient of Variation.
Lower percentage = Better precision.

SD: Standard Deviation (often calculated from the last 20 one-minute averages in steady-state).

$$CV = \frac{SD}{MEAN} \times 100\%$$

MEAN: Average concentration across the readings.



Practical Application (Reproducibility Analysis)

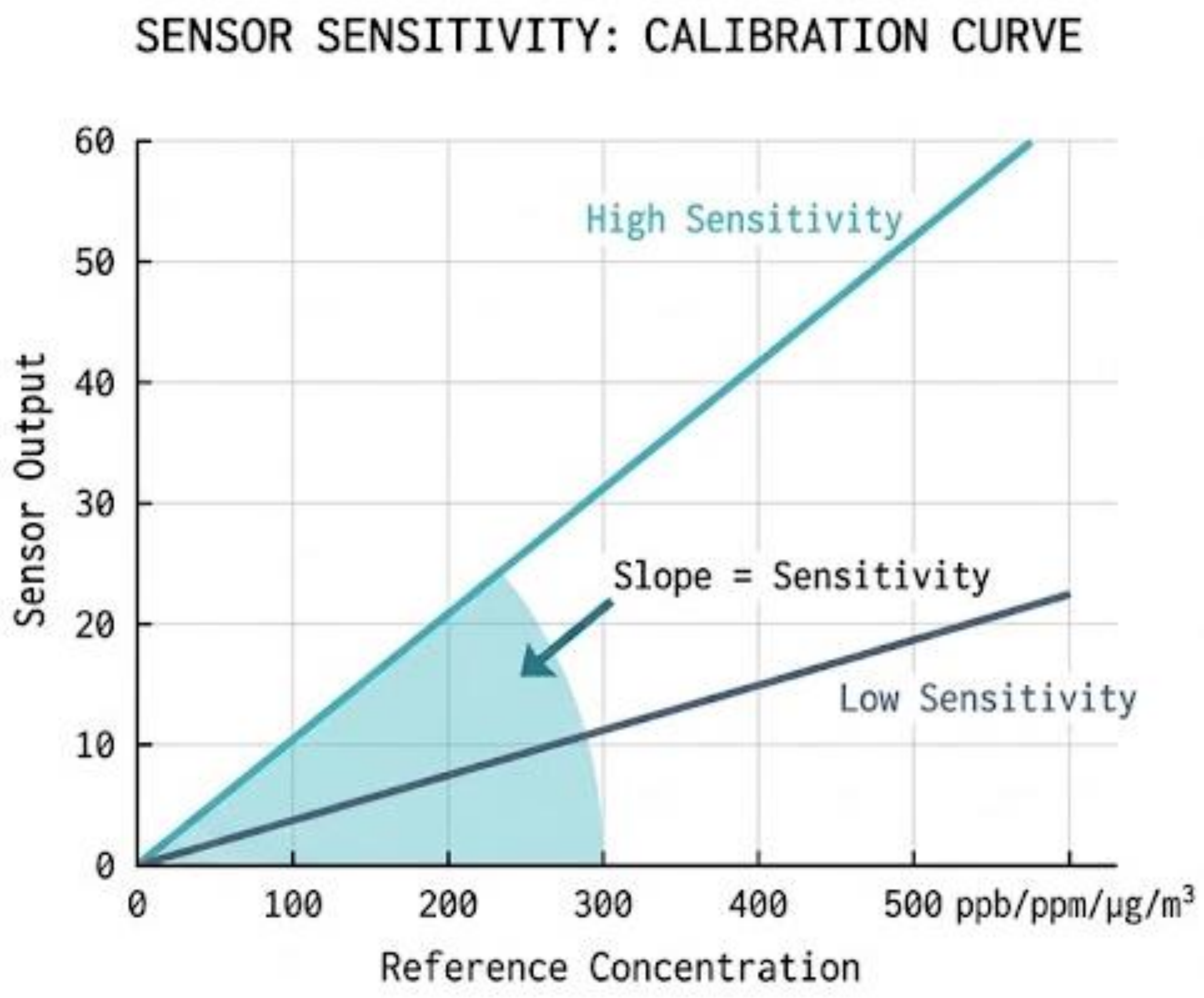
Because controlled environmental chambers are rare, precision is commonly measured by co-locating multiple sensors of the same kind in one location to test intra-model variability.

Target Standard

Regulatory protocols often target a CV <= 30% (and ideally intra-model variability <20% for triplicate sensors).

Funded by
the European Union

Sensitivity dictates a sensor's responsiveness to subtle pollutant changes



Definition
The sensor's response per unit change in pollutant concentration. It is the literal slope of the calibration curve.

Examples
Measured in nA per ppb/ppm for electrochemical gas sensors, or scattered light signal intensity per ug/m3 for optical PM sensors.

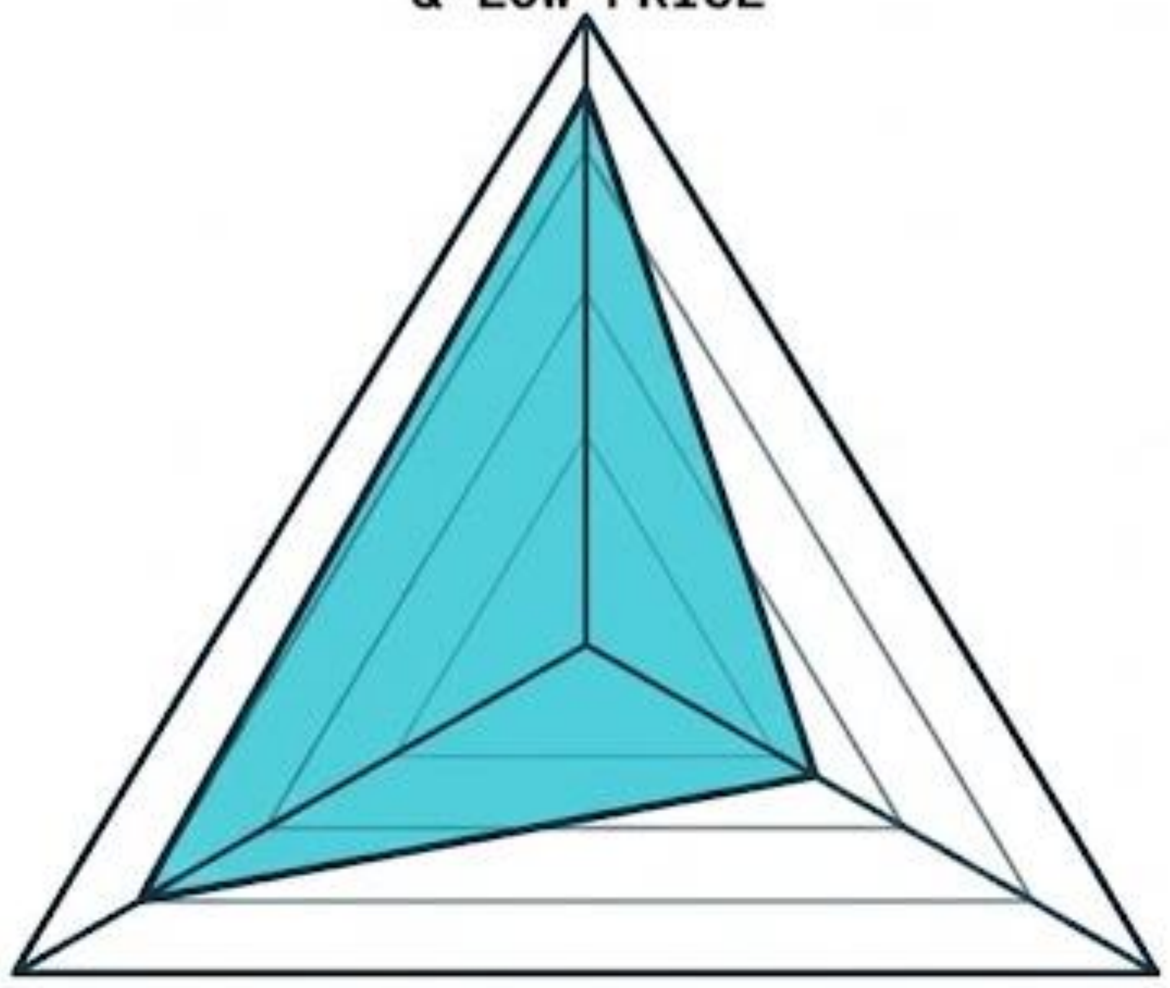
The Benefit
Higher sensitivity improves the ability to detect tiny atmospheric changes and directly lowers the Limit of Detection.

The Vulnerability
Sensitivity is highly susceptible to shifts in ambient air characteristics like temperature and relative humidity, leading to over- or under-estimations.

Funded by
the European Union

NextAIRE

Low-cost sensor design requires trading selectivity for miniaturization and price



THE OEM REALITY

Original Equipment Manufacturer (OEM) components used for detecting atmospheric gases and particles operate on strict design trade-offs.

CROSS-SENSITIVITY

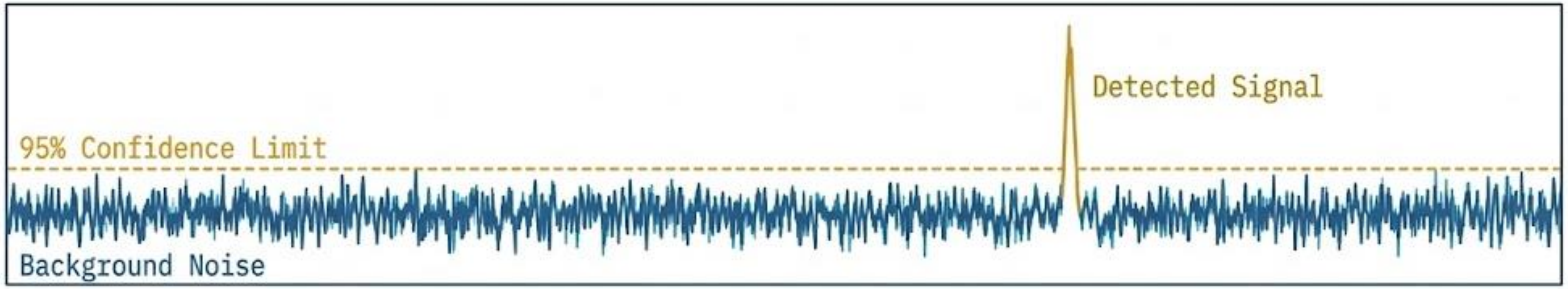
Electrochemical and metal oxide sensors intended for a specific gas often react to other gases in the environment (e.g., an NO2 sensor reacting to ambient O3).

MEASUREMENT ARTIFACTS

Compounding the selectivity issue, sensors exhibit measurement artifacts directly tied to temperature and humidity swings.

Funded by
the European Union

The limit of detection separates true environmental signals from baseline noise



$$\mathrm{LOD} = 3.3 \times \frac{\sigma}{S}$$

σ: Residual standard deviation (baseline noise).

S: Sensitivity (slope).

Definition

The lowest quantity of a pollutant distinguishable from the absence of that pollutant within a stated confidence limit.

Case Studies in Stability

A stable baseline is critical. The TGS-2611 chemiresistive CH_4 sensor achieves superior LOD primarily due to baseline stability.

Conversely, the MQ-4 sensor's 82 ppm LOD is a direct result of scatter in its calibration curve at low concentrations.

Funded by
the European Union

Additional statistical metrics provide a complete picture of sensor error

Repeatability

Consistency of readings under identical conditions.

Reproducibility

Consistency across different units of the exact same sensor model.

Saturation Limit

The absolute maximum concentration the sensor can reliably measure before the data flatlines.

MAE & RMSE

Mean Absolute Error and Root Mean Square Error. These measure the true magnitude of errors between sensor data and reference standards.

R-squared (Pearson Coefficient)

Highlights the correlation strength between sensor data and reference measurements. A high R-squared signifies strong predictive capability for tracking pollutant variations.

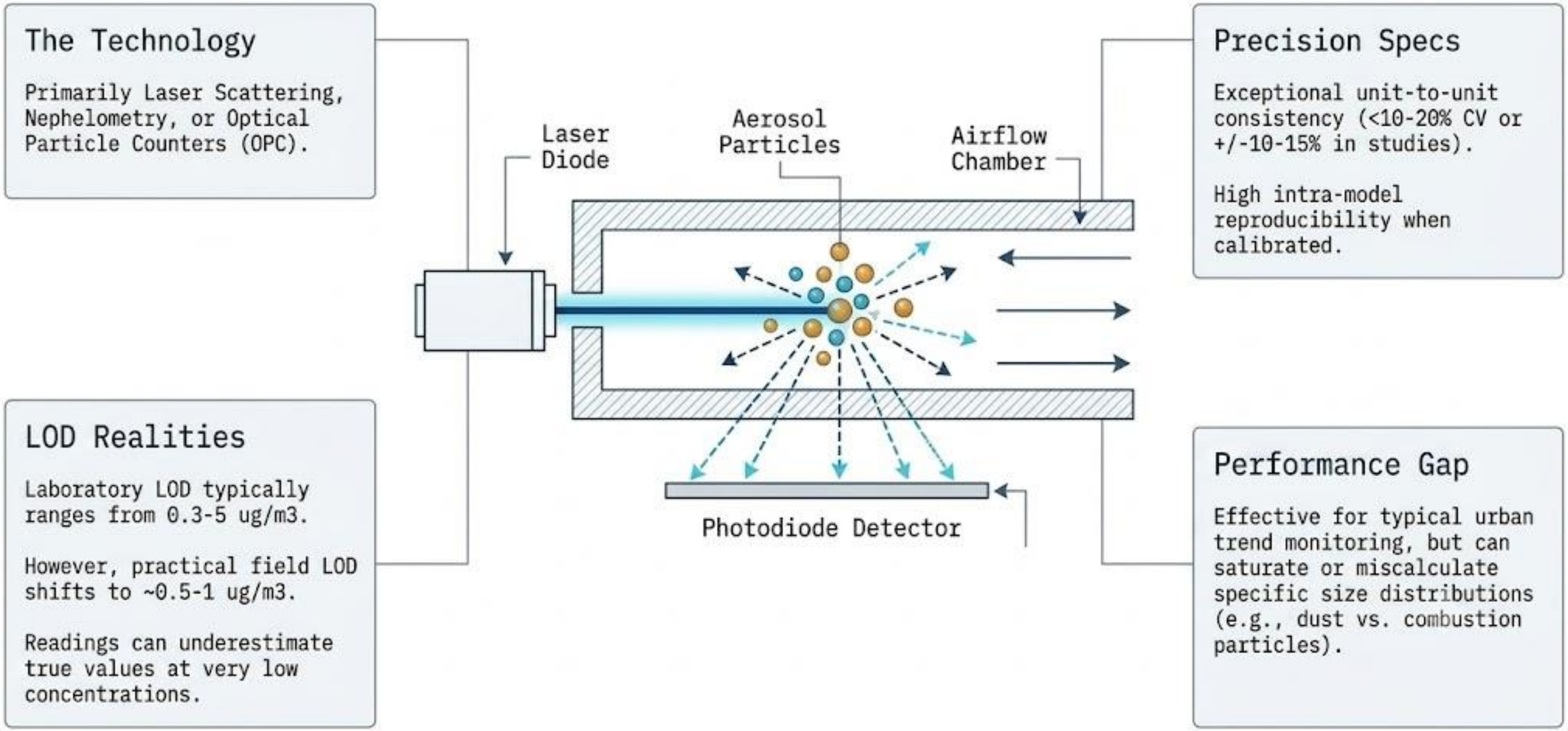
Funded by
the European Union

Different sensing mechanisms carry distinct performance profiles and vulnerabilities

	Optical / Laser (PM)	Electrochemical (NO_2, CO)	Metal-Oxide semiconductor (MOS)	NDIR (CO_2)
Mechanism	Light scattering / OPC	Generating electrons from chemical reactions	Chemiresistive reactions	Infrared light absorption
Advantage	Outperforms LEDs in accuracy; good intra-model reproducibility	High sensitivity (useful in safety devices)	Can reach sub-ppb in optimized versions	Excellent precision, low interference
Vulnerability	Degrades with relative humidity (RH) and complex aerosols	Cross-sensitivity (O_3 interference) and baseline shifts	Highly affected by humidity	Generally larger footprint and higher cost

Funded by
the European Union

Optical PM sensors



Funded by
the European Union

NextAIRE

Gas sensors

- NO_2 & O_3 (Electrochemical / MOS)

Lab LOD: Often <15 ppb

Field LOD: Spikes to 10-30 ppb due to noise and baseline shifts.

The Problem: Field R-squared drops to 0.1-0.7. Strong cross-sensitivity (O3 interfering with NO2) and low signal-to-noise ratios. Requires dual-sensor pairs and ML calibration outdoors.

- CO (Electrochemical)

Performance: Easier to track than NO2/O3 due to higher ambient levels (less critical LOD requirements). Exhibits good linearity.

- CO_2 (NDIR)

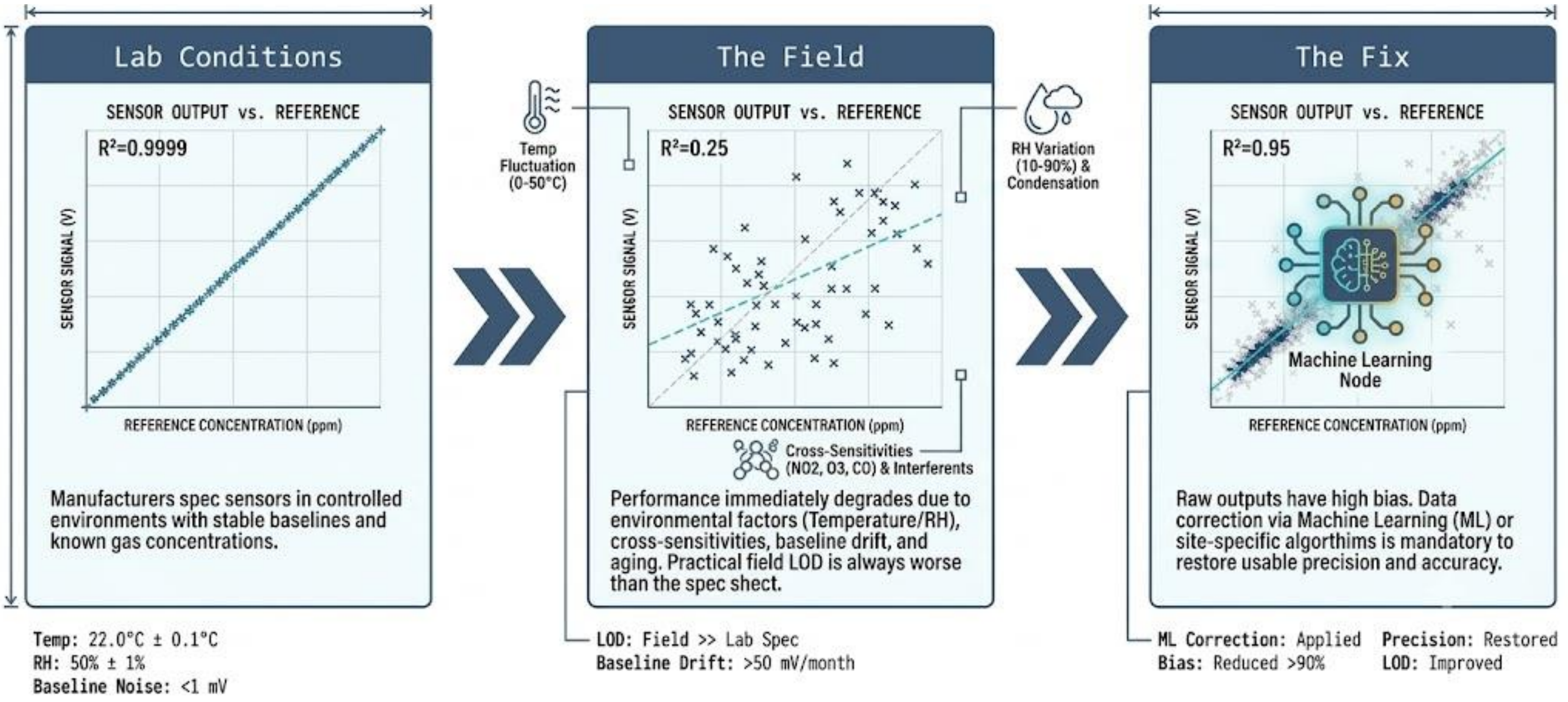
Performance: Excellent precision (+/-30 ppm or 3 ppm or ~3%), low LOD (<10-50 ppm), high linearity. Far less affected by cross-interferents.

Funded by
the European Union

NextAIRE

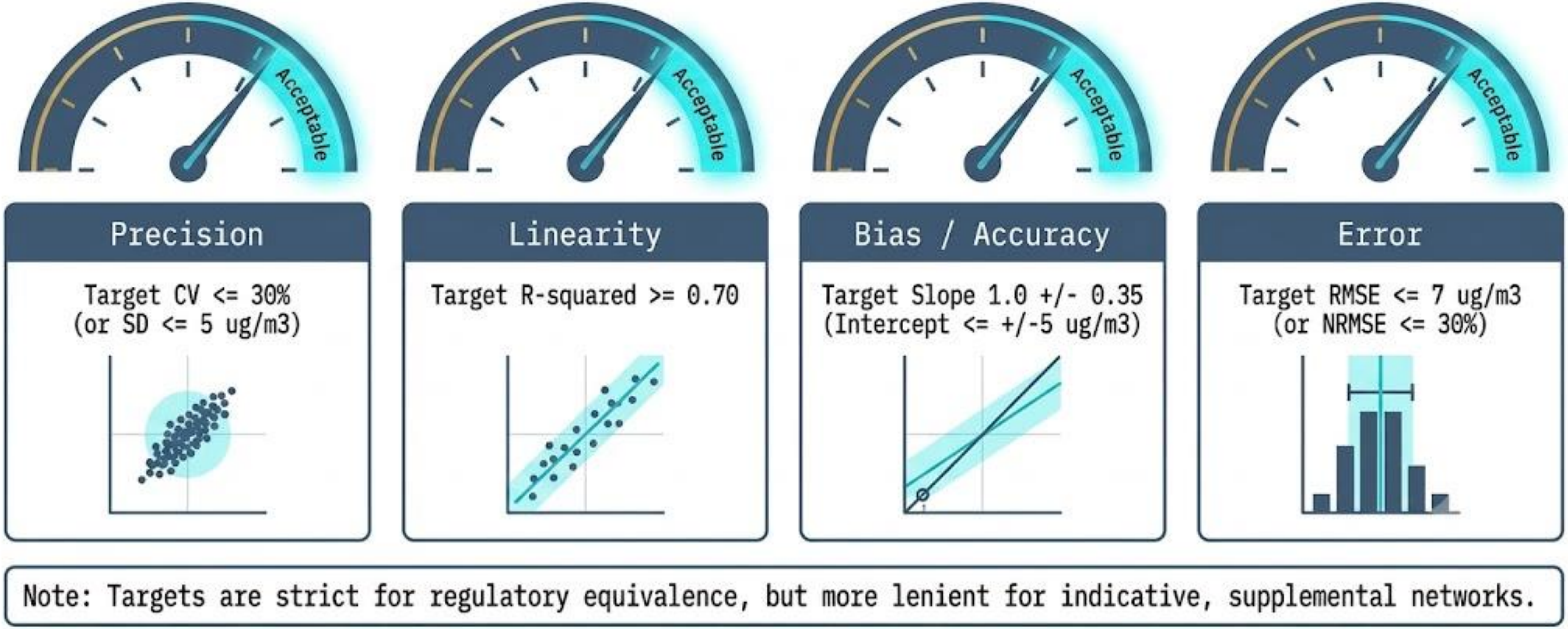
Real–worls deployment

(rapidly degrades pristine laboratoy performance specifications)



Funded by
the European Union

Regulatory frameworks



Funded by
the European Union

NextAIRE

Consortium

diadikasia BRIGHT GROUP MEMBER

smartsense

Megasense

CENTER FOR PHYSICAL SCIENCES AND TECHNOLOGY

INSTITUTE FOR ANTHROPOLOGICAL RESEARCH

theLisboncouncil think tank for the 21st century

INAR Institute for Atmospheric and Earth System Research

imi

msensis

vito

KNOW Center

UNIVERSITAS STUDIORUM SARAIEVOENSIS UNIVERZITET U SARAJEVU

ΔΗΜΟΚΡΙΤΟΣ DEMOKRITOS

FERIT FACULTY OF ELECTRICAL ENGINEERING, COMPUTER SCIENCE AND INFORMATION TECHNOLOGY OSIJEK

Funded by the European Union

Thank You!

Name: doc. dr.sc. Silvije Davila

Organization: Institute for Medical Research and Occupational Health

Email: sdavila@imi.hr

Website: www.imi.hr

NextAIRE

Funded by
the European Union